ACLEY BUJUNE

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Education

Dakota State University – GPA: 3.75/4.00 Madison, SD
Bachelor of Science in Computer Science Expected May 2027
Relevant Coursework: Advanced Data Structures & Algorithms, Foundations of Computation, Mathematical Foundations of AI, Deep Learning, Artificial Intelligence, Software Security, Parallel Computing, Algorithms and Optimization, Structure and System Analysis

Technical Skills

Python, TypeScript, JavaScript, Java, C++, C, SQL, Go, Rust, C#, HTML/CSS, React, Node.js, Spring Boot, Fastify, Bun, Kafka, Docker, Kubernetes, AWS (ECS, S3, RDS), Terraform, Git, Linux, SQLite, OpenTelemetry, Puppeteer, Zod, RESTful APIs, Microservices, Distributed Systems, System Design

Experience

Co-Founder & Software Engineer | Elevate Company Limited 2025 – Present
- Co-founded Elevate Company Limited, a software and technology consultancy delivering custom software, AI consulting, and data engineering solutions to clients across East Africa
- Architected Beam, a production-ready financial management platform for a construction company built with TypeScript, Bun, and SQLite — managing 3+ active projects, tracking $50,000+ in contractor payments, and delivering real-time profitability reporting across 7 core business domains
- Designed Beam's layered architecture with strict separation of UI, business logic, and data layers so that the platform can be white-labeled, extended with new modules, or customized for new industries without touching core business logic — cutting estimated onboarding time for new deployments by 50%

Software Engineering Intern | Cobros Logistics May 2025 – August 2025
- Developed a full-stack e-commerce logistics portal using React and Node.js so that Tanzanian customers could directly import goods from US retailers for the first time, processing $15,000+ in transactions
- Constructed a real-time price scraping engine using Puppeteer so that product costs from Amazon and Walmart were instantly converted and displayed in TZS, eliminating pricing errors and saving 20+ weekly admin hours
- Integrated localized payment gateways (M-Pesa, Tigo Pesa) via secure REST APIs so that unbanked users could complete transactions effortlessly, expanding the client's total addressable market by 35%

Projects

Tracedog (Causal Intelligence Engine) | TypeScript, Bun, React, Fastify, SQLite, OpenTelemetry
- Designed a deterministic causal analysis engine parsing OpenTelemetry span trees depth-first to classify each error as root cause or propagated symptom — reducing mean time to root cause from hours to seconds across distributed systems with 0.95 base confidence scoring
- Implemented a feedback-loop memory layer in SQLite so that confirmed failure patterns graduate to learned rules over time, cutting repeat-incident diagnosis time by 70% by eliminating redundant graph traversal on known failure signatures
- Constrained Claude Sonnet to a pure narration layer that translates engine conclusions into plain-English incident reports so that on-call engineers receive actionable root cause summaries without sacrificing the engine's full determinism
- Deployed a Fastify REST API with Zod-validated schemas supporting 3 ingestion modes — JSON upload, file import, and direct OTel Collector export — so that SREs could integrate TraceRoot into any existing observability pipeline without changing their infrastructure

C Compiler (Systems Programming) | C++, Make, Linux, Assembly Aug 2025 – Nov 2025
- Built a fully functional C compiler from scratch in C++17 implementing all 5 compilation stages — lexing, parsing, AST generation, code generation, and assembly — so that any C source file could be compiled to binary without external tooling
- Developed an interactive React frontend visualizing each compilation stage in real time so that users could inspect lexer tokens, AST nodes, generated code, and final assembly output step-by-step, making compiler internals accessible to anyone
- Applied recursive descent parsing techniques developed for this compiler to architect Tracedog's span tree traversal engine, enabling depth-first causal analysis across distributed traces with deterministic, reproducible results

Competitive Chess Engine (NanoCon Winner) | Python, PyTorch, Bitboards Sept 2025 – Nov 2025
- Won 1st place out of 30+ teams at NanoCon 2025 Hackathon by building a high-performance chess engine in Python that outplayed every competing bot through superior search depth and evaluation accuracy
- Optimized move generation using Bitboard representation and Alpha-Beta pruning so that the engine searched 20+ moves deep in seconds, dramatically reducing the search space without sacrificing tactical accuracy
- Trained a PyTorch neural network to evaluate board positions so that the engine could assess complex middlegame and endgame positions beyond what handcrafted heuristics could capture, improving tactical decision-making against opponent bots

Leadership & Extracurricular

Honors: Beadle Honors Program — Selected for academic excellence, Dakota State University, maintaining 3.75 GPA across a rigorous CS curriculum
Hackathon: NanoCon 2025 — 1st Place winner among 30+ teams for best AI/systems project
Black Students Association: Officer — Leading community outreach and campus diversity initiatives for 200+ student body
Member: AI Club, Computer Club — Regular competitor in technical hackathons and coding workshops